Therefore, The expected frequency at 0 is

$$N \times P(0)$$
$$= 100 \times 0.4066$$
$$= 40.66$$

We use recurrence formula to find the other expected frequencies.

The expected frequency at $x+1$ is

$$\frac{m}{x+1} \times \text{Expected frequency at } x$$

x	$\frac{m}{x+1}$	Expected frequency at $x = N\,P(x)$
0	0.9	40.66
1	$\frac{0.9}{2}$	36.594
2	$\frac{0.9}{3}$	16.4673
3	$\frac{0.9}{4}$	4.94019
4	$\frac{0.9}{5}$	1.1115
5	$\frac{0.9}{6}$	0.20007

Table of expected frequency distribution (on rounding to the nearest integer)

x	0	1	2	3	4	5
Expected frequency	41	37	16	5	1	0

Step 3 : Level of significance

$\alpha = 5\%$

Step 4 : Test statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

12th_Statistics_EM_Unit_2.indd 66 1/28/2020 7:10:11 PM